

examples

February 14, 2026

1 twowaypanel — Examples Gallery (Notebook)

This notebook contains the executable code for the **twowaypanel Examples Gallery** in the online documentation:

- <https://twowaypanel.readthedocs.io/en/latest/examples/examples.html>

For tutorials and the full API reference, see the documentation homepage:

- <https://twowaypanel.readthedocs.io/>

2 Import and setup

```
[1]: import numpy as np
import twowaypanel
```

3 Example 1: Binary logit with strictly exogenous regressors (Angrist and Evans 1998)

```
[2]: data = twowaypanel.database.angristevans98()

# Typical workflow (exact variable names depend on the shipped dataset):
data = data.sort_values(["id", "year"])
N = data["id"].nunique()
T = data["year"].nunique()

# Y: LFP (N x T)
lfp = data["lfp"].to_numpy().reshape(N, T)

# X: covariates (N x T x K)
X_static = data.loc[:, ['kids0_2', 'kids3_5', 'kids6_17', 'ln_husincome']].
    to_numpy().reshape(N, T, 4)
X_dynamic = data.loc[:, ['lag_lfp', 'kids0_2', 'kids3_5', 'kids6_17', 'ln_husincome']].
    to_numpy().
    reshape(N, T, 5)
X_static_names = ["Kids 0-2", "Kids 3-5", "Kids 6-17", "ln(hus inc)"]
```

```
X_dynamic_names = ["LFP_t-1", "Kids 0-2", "Kids 3-5", "Kids 6-17", "ln(hus_
↳inc)"]
```

Joint MLE (no bias correction)

```
[3]: res_mle = twowaypanel.fit(
    lfp, X_static,
    model="logit", prior=None,
    algorithm="JML",
    X_names=X_static_names
)
```

----- ESTIMATION RESULTS -----

LOGIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
WITHOUT BIAS CORRECTION

```
Number of Individuals: 664      Observations: 5976
Number of Time Periods: 9      Log-likelihood: -3033.74
Algorithm: Joint MLE          Time spent (seconds): 1.098
```

Independent variable	Coefficient	Std. Err.	P> z	[95% conf. interval]	
Kids 0-2	-1.17434565	0.09836036	0.000	-1.36712	-0.98156
Kids 3-5	-0.59134501	0.08622960	0.000	-0.76034	-0.42234
Kids 6-17	-0.01566283	0.06075953	0.797	-0.13474	0.103419
ln(hus inc)	-0.40458145	0.09432568	0.000	-0.58945	-0.21971

Note: Panel data contains under-identified observations;
Dropped 797 out of 1461 individuals, and 0 out of 9 time periods.

4 Example 2: Binary logit with strictly exogenous regressors + model-specific prior (MCMC) + convergence diagnostics

```
[4]: beta_variance = np.array([np.sqrt(1e-1),np.sqrt(1e-2),np.sqrt(1e-2),np.
    ↳sqrt(5e-5),np.sqrt(5e-7)])

res_mcmc = twowaypanel.fit(
    lfp, X_static,
    model="logit", prior="SE", algorithm="MCMC",
    X_names=X_static_names,
    beta_variance=beta_variance,fe_variance=0.1,
    mcmc_iters=150000, mcmc_burnin=50000, mcmc_skipsize=20,
    mcmc_timer=1, mcmc_sv_mle=True, mcmc_diagnosis=True
)
```

100%|
150000/150000 [59:00:49<00:00, 1.42s/it]

----- ESTIMATION RESULTS -----

LOGIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
USING BIAS-REDUCING PRIOR FOR EXPONENTIAL FAMILY MODELS
WITH STRICTLY EXOGENOUS REGRESSORS

Number of Individuals: 664 Observations: 5976
Number of Time Periods: 9
Algorithm: Markov Chain Monte Carlo Time spent (minutes): 3540.872
150000 repetitions

Independent variable	Coefficient	Std. Err.	P> z	[95% conf. interval]	
Kids 0-2	-0.96974154	0.09139083	0.000	-1.14885	-0.79062
Kids 3-5	-0.48227300	0.08078112	0.000	-0.64059	-0.32395
Kids 6-17	0.001101959	0.05723464	0.985	-0.11107	0.113276
ln(hus inc)	-0.37748241	0.08895669	0.000	-0.55182	-0.20313

----- AVERAGE PARTIAL EFFECTS -----

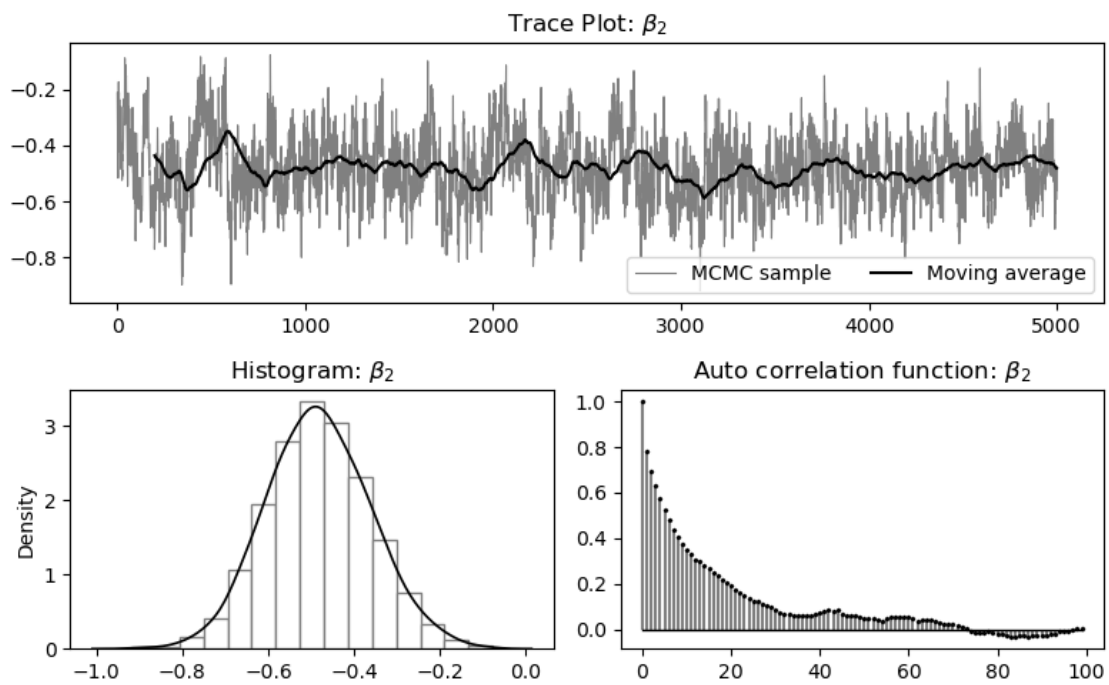
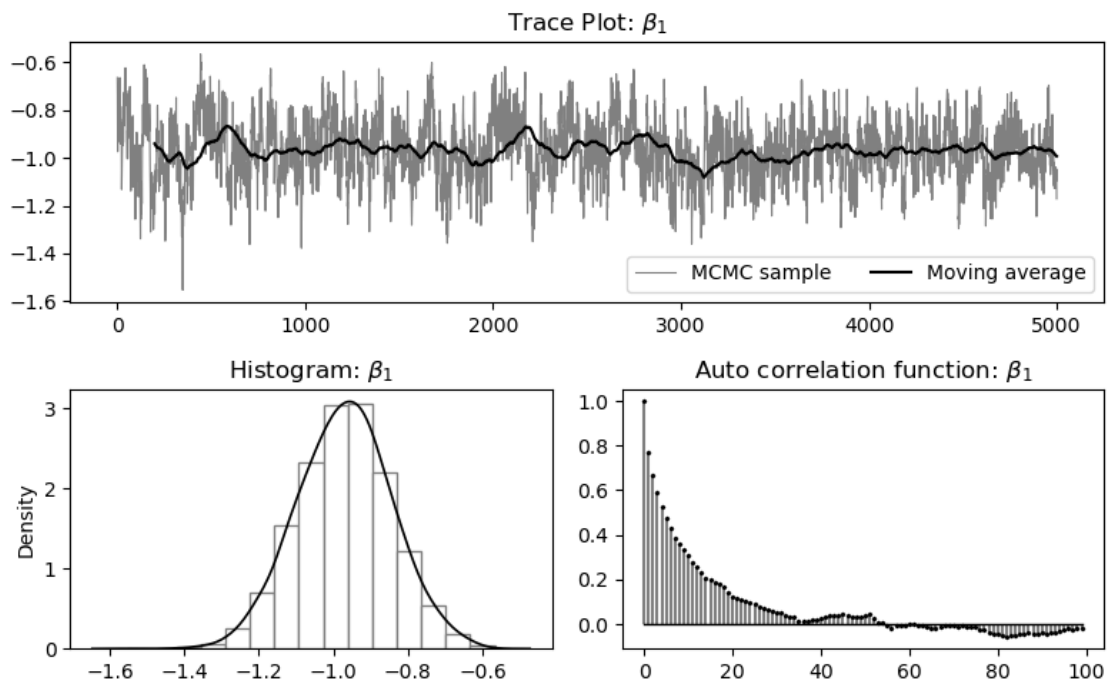
Independent variable	Coefficient	Std. Err.	P> z	[95% conf. interval]	
Kids 0-2	-0.09161824	0.03829473	0.017	-0.16667	-0.01656
Kids 3-5	-0.04556369	0.01987691	0.022	-0.08452	-0.00660
Kids 6-17	0.000104109	0.00487730	0.983	-0.00945	0.009663
ln(hus inc)	-0.03566339	0.01644653	0.030	-0.06789	-0.00342

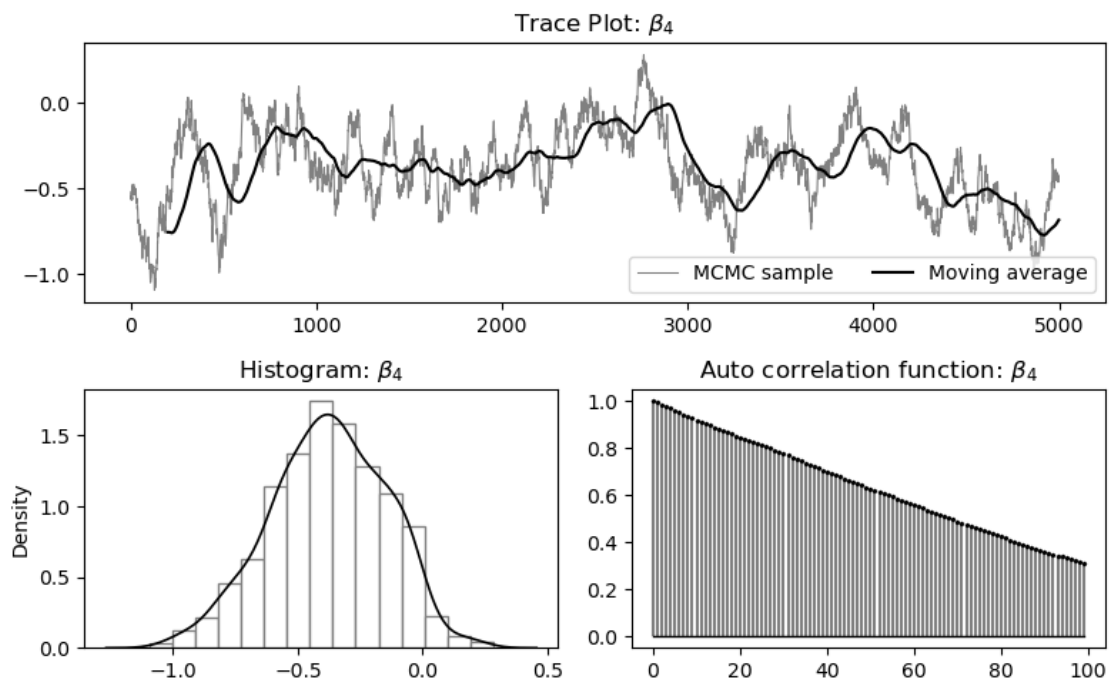
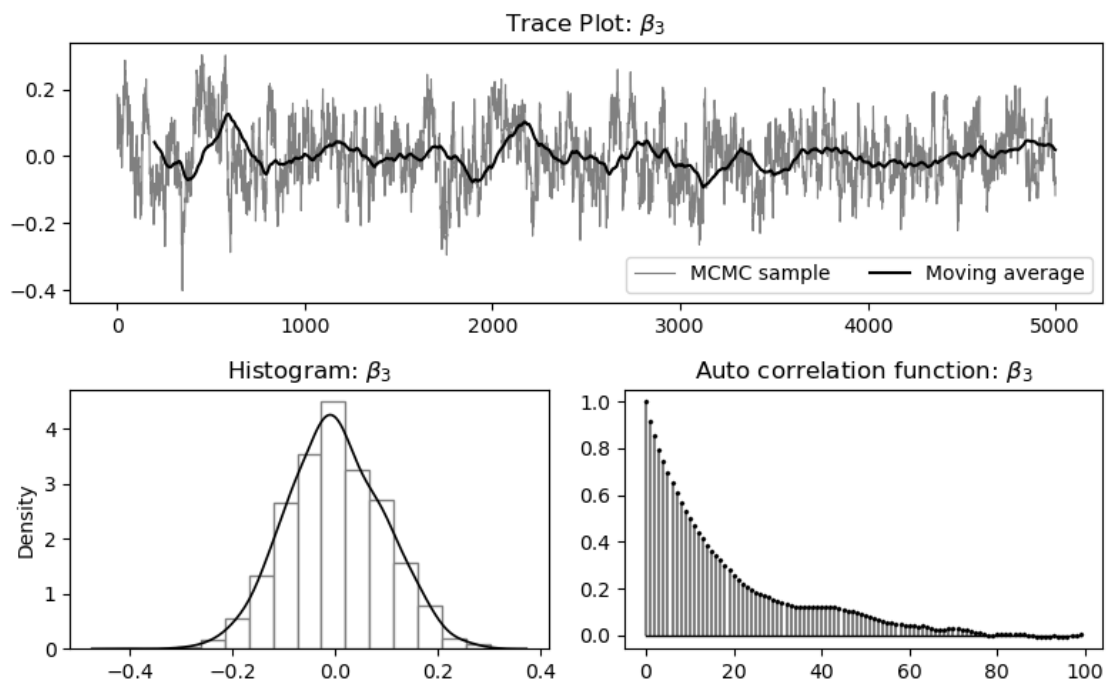
Note: Panel data contains under-identified observations;
Dropped 797 out of 1461 individuals, and 0 out of 9 time periods.

----- MCMC DIAGNOSIS -----

Geweke (1992) Convergence Diagnostic Test
First 10% sample vs. 20 segments
of final 50% sample

Independent variable	Acceptance(%)	Average p-value	Smallest p-value
Kids 0-2	59.202592%	0.886	0.744
Kids 3-5	57.159571%	0.855	0.682
Kids 6-17	58.565585%	0.852	0.689
ln(hus inc)	30.478304%	0.573	0.161





5 Example 3: Analytical bias correction (Fernández-Val and Weidner, 2016)

```
[5]: res_fw16 = twowaypanel.fit(
    lfp, X_static,
    model="logit", prior=None, ac=True,
    algorithm="JML",
    X_names=X_static_names
)
```

```
-----
---- ESTIMATION RESULTS -----
                        LOGIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
                        ANALYTICAL BIAS CORRECTION ON ESTIMATOR (FW16)
-----
Number of Individuals:   664                      Observations:      5976
Number of Time Periods:    9                      Log-likelihood: -3033.74
Algorithm: Joint MLE                      Time spent (seconds):    2.425
-----
```

Independent variable	Coefficient	Std. Err.	P> z	[95% conf. interval]
Kids 0-2	-1.02689349	0.09836036	0.000	-1.21966 -0.83411
Kids 3-5	-0.51776197	0.08622960	0.000	-0.68676 -0.34876
Kids 6-17	-0.01343869	0.06075953	0.825	-0.13252 0.105643
ln(hus inc)	-0.35653581	0.09432568	0.000	-0.54140 -0.17166

```
-----
Note: Panel data contains under-identified observations;
      Dropped 797 out of 1461 individuals, and 0 out of 9 time periods.
```

6 Example 4: Dynamic probit with generic prior

```
[6]: res_dyn = twowaypanel.fit(
    lfp, X_dynamic,
    model="probit", prior="Generic", lag=1,
    algorithm="JML",
    X_names=X_dynamic_names
)
```

```
-----
---- ESTIMATION RESULTS -----
                        PROBIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
```

USING BIAS-REDUCING PRIOR (GENERIC VERSION)

```

Number of Individuals: 664                      Observations: 5976
Number of Time Periods: 9                      Log-likelihood: -3148.29
Algorithm: Joint MLE                          Time spent (seconds): 1.800

```

Independent variable	Coefficient	Std. Err.	P> z	[95% conf. interval]	
LFP_t-1	0.970193330	0.04268741	0.000	0.886530	1.053856
Kids 0-2	-0.46261536	0.05690305	0.000	-0.57413	-0.35109
Kids 3-5	-0.17731722	0.05122717	0.001	-0.27771	-0.07691
Kids 6-17	0.036581388	0.03617607	0.312	-0.03432	0.107482
ln(hus inc)	-0.22587568	0.05479406	0.000	-0.33326	-0.11848

----- AVERAGE PARTIAL EFFECTS -----

Independent variable	Coefficient	Std. Err.	P> z	[95% conf. interval]	
LFP_t-1	0.166942566	0.00717625	0.000	0.152877	0.181007
Kids 0-2	-0.06130859	0.02691777	0.023	-0.11406	-0.00855
Kids 3-5	-0.02349915	0.01179089	0.046	-0.04660	-0.00039
Kids 6-17	0.004847987	0.00490677	0.323	-0.00476	0.014464
ln(hus inc)	-0.02993441	0.01437215	0.037	-0.05810	-0.00176

Note: The trimming parameter for estimating spectral expectation is 1.

Note: Panel data contains under-identified observations;

Dropped 797 out of 1461 individuals, and 0 out of 9 time periods.

7 Example 5: Ordered logit (simulated data)

```

[7]: # Generate a dynamic ordered-logit panel (returns long-form arrays by default)
FEi, FEt, FE, index_i, index_t, Y, X, Ystar, alphas0, gammas0 = twowaypanel.
demo.PanelGenData(
    N=45, T=15, seed=10, model="ologit", dynamic=1
)

# Reshape into N x T and N x T x K
Y = Y.reshape(45, 15)
X = X.reshape(45, 15, 2)

```

```

[8]: res_ologit_mle = twowaypanel.fit(
    Y, X, model="ologit",
    prior=None, algorithm="JML",

```

```
cutoff0=-2.5,      # identification normalization
)
```

----- ESTIMATION RESULTS -----

ORDERED LOGIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
WITHOUT BIAS CORRECTION

Number of Individuals:	45	Observations:	675
Number of Time Periods:	15	Log-likelihood:	-685.19
Algorithm: Joint MLE		Time spent (seconds):	0.096

	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	0.441343774	0.18110680	0.015	0.086392	0.796295
X2	1.241542606	0.11763947	0.000	1.010980	1.472104
/cutoff2	0.466886524	0.17628462	0.008	0.121386	0.812386
/cutoff3	2.698815569	0.22275833	0.000	2.262231	3.135399

----- AVERAGE PARTIAL EFFECTS -----

Pr(Y=1)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	0.030253733	0.01309578	0.021	0.004587	0.055920
X2	0.084568510	0.01406633	0.000	0.056999	0.112137
Pr(Y=2)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	0.048848728	0.02139962	0.022	0.006907	0.090789
X2	0.131727845	0.01749094	0.000	0.097447	0.166008
Pr(Y=3)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	-0.03211661	0.01563286	0.040	-0.06275	-0.00147
X2	-0.08122116	0.01867050	0.000	-0.11781	-0.04462
Pr(Y=4)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	-0.04698584	0.01937041	0.015	-0.08494	-0.00902
X2	-0.13507518	0.01714204	0.000	-0.16867	-0.10147


```
[9]: res_ologit_generic = twowaypanel.fit(
    Y, X, model="ologit",
    prior="Generic", algorithm="JML", lag=1,
    cutoff0=-2.5,
)
```

----- ESTIMATION RESULTS -----

ORDERED LOGIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
USING BIAS-REDUCING PRIOR (GENERIC VERSION)

Number of Individuals:	45	Observations:	675
Number of Time Periods:	15	Log-likelihood:	-711.48
Algorithm: Joint MLE		Time spent (seconds):	0.116

	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	0.571509060	0.17854978	0.001	0.221569	0.921448
X2	1.169471961	0.11501164	0.000	0.944060	1.394883
/cutoff2	0.309800227	0.16596220	0.062	-0.01546	0.635069
/cutoff3	2.427988601	0.20935639	0.000	2.017671	2.838306

----- AVERAGE PARTIAL EFFECTS -----

Pr(Y=1)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	0.041244510	0.01443204	0.004	0.012959	0.069529
X2	0.083908563	0.01388271	0.000	0.056699	0.111117
Pr(Y=2)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	0.068221113	0.02150758	0.002	0.026068	0.110373
X2	0.131137738	0.01657559	0.000	0.098651	0.163624
Pr(Y=3)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	-0.04623204	0.01652339	0.005	-0.07861	-0.01384
X2	-0.08136355	0.01729637	0.000	-0.11526	-0.04746
Pr(Y=4)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
X1	-0.06323357	0.02011343	0.002	-0.10265	-0.02381
X2	-0.13368274	0.01678849	0.000	-0.16658	-0.10077

Note: The trimming parameter for estimating spectral expectation is 1.

```
[10]: res_ologit_mcmc = twowaypanel.fit(
      Y, X, model="ologit",
      prior="Generic", algorithm="MCMC", lag=1,
      cutoff0=-2.5,
      mcmc_iters=120000, mcmc_burnin=20000, mcmc_skipsize=20, mcmc_diagnosis=True
    )
```

```
100%|
120000/120000 [2:57:41<00:00, 11.26it/s]
```

----- ESTIMATION RESULTS -----

ORDERED LOGIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
USING BIAS-REDUCING PRIOR (GENERIC VERSION)

```
Number of Individuals:    45                      Observations:    225
Number of Time Periods:   15
Algorithm: Markov Chain Monte Carlo                Time spent (minutes): 177.691
120000 reptitions
```

	Coefficient	Std. Err.	P> z	[95% conf. interval]
X1	0.568049498	0.17823331	0.001	0.218730 0.917368
X2	1.167454158	0.11479293	0.000	0.942471 1.392436
/cutoff1	0.292924966	0.16484033	0.076	-0.03014 0.615995
/cutoff2	2.400585164	0.20798825	0.000	1.992948 2.808221

----- AVERAGE PARTIAL EFFECTS -----

Pr(Y=0)	Coefficient	Std. Err.	P> z	[95% conf. interval]
X1	0.041390922	0.01451946	0.004	0.012934 0.069847
X2	0.084540492	0.01391589	0.000	0.057266 0.111814

Pr(Y=1)	Coefficient	Std. Err.	P> z	[95% conf. interval]
X1	0.067626969	0.02139310	0.002	0.025698 0.109555
X2	0.130605235	0.01649740	0.000	0.098271 0.162938

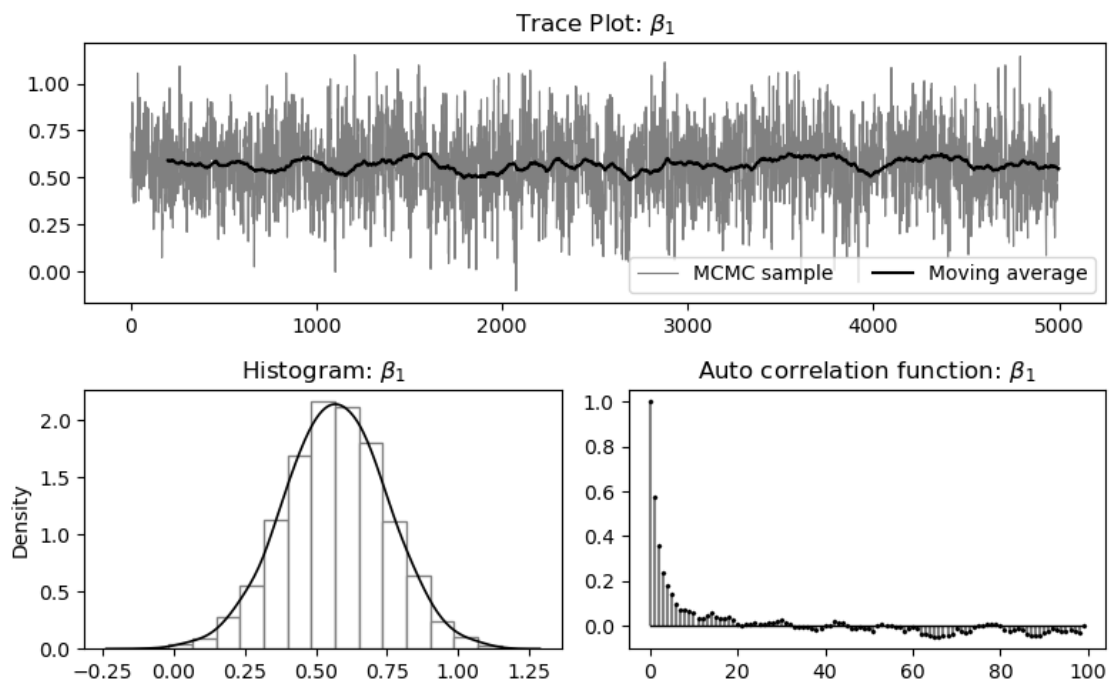
Pr(Y=2)	Coefficient	Std. Err.	P> z	[95% conf. interval]
X1	-0.04584389	0.01643358	0.005	-0.07805 -0.01363

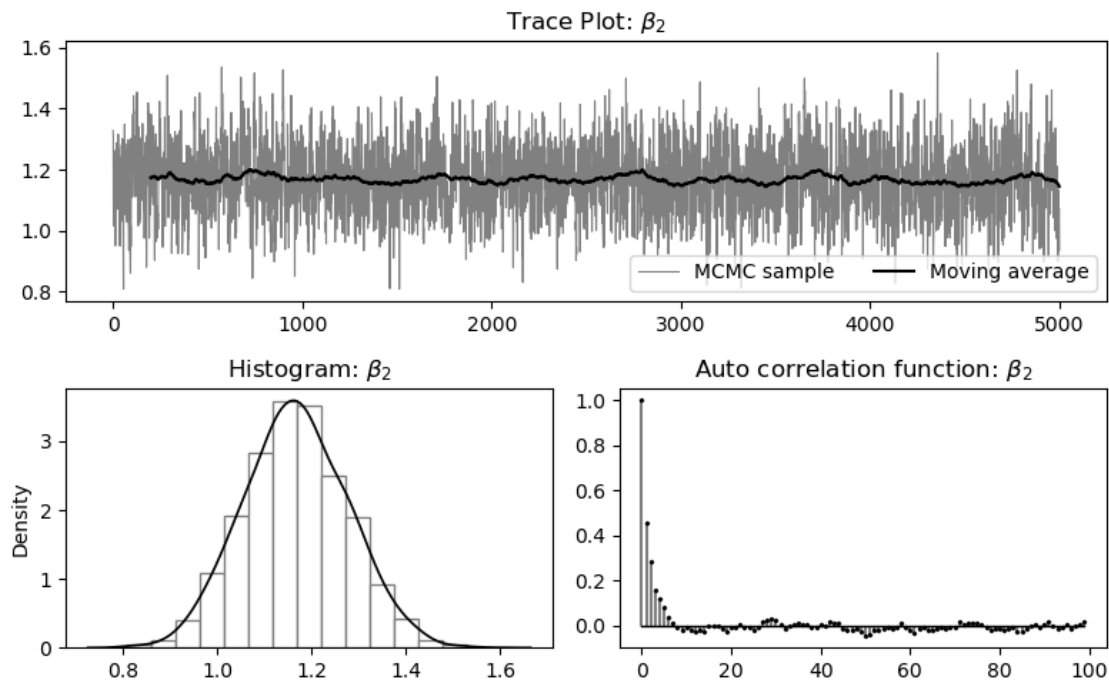
	X2	-0.08104455	0.01721884	0.000	-0.11479	-0.04729
	Pr(Y=3)	Coefficient	Std. Err.	P> z	[95% conf. interval]	
	X1	-0.06317400	0.02016498	0.002	-0.10269	-0.02365
	X2	-0.13410117	0.01678597	0.000	-0.16699	-0.10120

Note: The trimming parameter for estimating spectral expectation is 1.

----- MCMC DIAGNOSIS -----

Geweke (1992) Convergence Diagnostic Test First 10% sample vs. 20 segments of final 50% sample			
Independent variable	Acceptance(%)	Average p-value	Smallest p-value
X1	42.594425%	0.914	0.799
X2	41.867418%	0.951	0.882
/cutoff1	36.887368%	0.942	0.781
/cutoff2	59.267592%	0.938	0.770





8 Example 6: Multinomial logit (simulated data)

```
[11]: _, _, _, _, _, Y, X, _, _, _ = twowaypanel.demo.PanelGenData(
    N=45, T=15, seed=10, model="mlogit", dynamic=1
)
Y = Y.reshape(45, 15)
X = X.reshape(45, 15, 2)

# Uncorrected JML
res_mlogit_mle = twowaypanel.fit(
    Y, X, model="mlogit",
    prior=None, algorithm="JML"
)
```

 ESTIMATION RESULTS

MULTINOMIAL LOGIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
 WITHOUT BIAS CORRECTION

Number of Individuals:	44	Observations:	660
Number of Time Periods:	15	Log-likelihood:	-590.37
Algorithm: Joint MLE		Time spent (seconds):	0.810

Independent variable		Coefficient	Std. Err.	P> z	[95% conf. interval]	

Y = 2						
	X1	-0.02201089	0.25903835	0.932	-0.52970	0.485678
	X2	1.006845054	0.17222417	0.000	0.669302	1.344387
Y = 3						
	X1	0.797433329	0.26237748	0.002	0.283199	1.311666
	X2	1.286031037	0.17948933	0.000	0.934249	1.637812

----- AVERAGE PARTIAL EFFECTS -----

Independent variable		Coefficient	Std. Err.	P> z	[95% conf. interval]	

Y = 2						
	X1	0.026330907	0.01542943	0.088	-0.00390	0.056571
	X2	0.043227913	0.02556975	0.091	-0.00688	0.093342
Y = 3						
	X1	0.028183105	0.01622722	0.082	-0.00362	0.059986
	X2	0.120234102	0.02433605	0.000	0.072537	0.167930

Note: Panel data contains under-identified observations;
Dropped 1 out of 45 individuals, and 0 out of 15 time periods.

```
[12]: res_mlogit_dml = twowaypanel.fit(
    Y, X,
    model="mlogit", prior="PML", lag=1,
    algorithm="JML"
)
```

----- ESTIMATION RESULTS -----

MULTINOMIAL LOGIT PANEL MODEL WITH TWO-WAY FIXED EFFECTS
USING BIAS-REDUCING PRIOR FOR EXPONENTIAL FAMILY MODELS
WITH PREDETERMINED REGRESSORS

Number of Individuals:	44	Observations:	660
Number of Time Periods:	15	Log-likelihood:	-520.18
Algorithm: Joint MLE		Time spent (seconds):	4.762

Independent variable	Coefficient	Std. Err.	P> z	[95% conf. interval]	

Y = 2					
X1	0.284336843	0.25121981	0.258	-0.20802	0.776702

Y = 3	X2	0.926801399	0.16670645	0.000	0.600073	1.253529
	X1	0.752554189	0.25684368	0.003	0.249166	1.255942
	X2	1.190567767	0.17313990	0.000	0.851230	1.529904

----- AVERAGE PARTIAL EFFECTS -----

Independent variable		Coefficient	Std. Err.	P> z	[95% conf. interval]	

Y = 2						
	X1	0.038674882	0.01617262	0.017	0.006978	0.070371
	X2	0.041445205	0.02556205	0.105	-0.00865	0.091544
Y = 3						
	X1	0.040827271	0.01709489	0.017	0.007322	0.074331
	X2	0.124371599	0.02436747	0.000	0.076613	0.172129

Note: The trimming parameter for estimating spectral expectation is 1.

Note: Panel data contains under-identified observations;

Dropped 1 out of 45 individuals, and 0 out of 15 time periods.